Instrument list

I/O count

control-valve check

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Teaching document. Numbers come from the MEA CO2 Capture Ver 1 simulator model (design point, noise off); items marked TBC/assumed are not design data. No vendor, client or contractor data. Checker and approver are left blank on purpose.

Contents

1 Purpose, scope and basis . . . 3
1.1 References . . . 3
1.2 How the rows were made . . . 3
1.3 Assumptions . . . 3
2 Instrument list . . . 4
3 Final elements and motors . . . 6
4 I/O count . . . 6
5 Control-valve check . . . 7
5.1 Worked calculation · FV-201 (liquid, IEC 60534-2-1) . . . 7
6 Alarm and trip hierarchy . . . 7
7 Where the trips and SIFs are specified . . . 7
8 Engineering queries . . . 8

1 Purpose, scope and basis

This instrument list registers every instrument of the MEA CO2 capture unit: the 40 bubbles of the screen P&ID and the indicators of the DCS (44 rows), the final elements and motors, the DCS and SIS I/O count, and a control-valve sizing check. For each instrument it gives the service, the field devices with their I/O type and system, a typical instrument type, the calibrated range, the normal value, the alarm and trip settings, the fail position, the SIL and the P&ID reference.

Teaching document. Numbers come from the MEA CO2 Capture Ver 1 simulator model (design point, noise off); items marked TBC/assumed are not design data.

1.1 References

- CC-PID-101 rev A (P&ID)
- MEA-PFD-001 rev 0 (PFD)
- MEA-IN-CE-001 rev 0 (C&E matrix: what each trip and alarm does)
- MEA-IN-SRS-001 rev 0 (SRS: SIL, voting, proof test of each SIF)
- MEA-PE-RLB-001 rev 0 (relief load basis: PSV-301 set)
- ISA-5.1 (identification letters)
- IEC 60534-2-1 (control-valve sizing equations)

1.2 How the rows were made

The rows are generated from the simulator's own tables, so the list and the OTS cannot drift apart. The screen bubbles give the tag and the display. The indicators give the range and service, and the controllers give the span, the final element and its fail position. The alarm list gives the alarm limits and the trip constants give the trip settings. The normal value is the design state with noise off. Field device tags follow the ISA-5.1 letters: loop FIC-201 → transmitter FT-201 (AI) and valve FV-201 (AO). The trip switches become separate SIS transmitters (B suffix) because the model trips on its own sensors. The SIL column is copied from MEA-IN-SRS-001.

1.3 Assumptions

- Control-valve pressure drops (FV-109/201/303 100 kPa, TV-202/110/302 70 kPa, FV-207 50 kPa, FV-208 150 kPa) are assumed: the model has no valve ΔP except PV-301 (175 → 150 kPa a, K.Pdown).
- Valve characteristic in the model is linear (flow = capacity × opening); the equal-percentage opening (R = 50) is shown for comparison.
- Gas sizing (IEC 60534-2-1): x_T 0.70 (globe, typical), k 1.33 steam / 1.29 CO2, ideal-gas density; the steam density is at 3.5 bar a saturated.
- Instrument types are typical selections (TBC by the instrument engineer), not vendor data.
- SIS transmitters LT-109B, LT-208B, PT-301B and TT-206B are the model's own trip sensors (sim.meas.LS109, LS208, PS301, TS206); the B suffix is this list's numbering.
- I/O count: one shared SIS transmitter LT-109B serves LSHH-109 and LSLL-109; spares 20 % (typical practice) are added in the summary.

2 Instrument list

44 rows: 40 screen bubbles in drawing order, then the indicators that have no bubble. Rows with a trip are SIS instruments.

Tag	Service	Field device · I/O · system	Type (typical, TBC)	Range	Units	Normal	Alarm	Trip	Fail position	SIL	P&ID
FIC-101	Flue gas to T-101 → blower K-101 speed (VFD)	FT-101 · AI · DCS SC-101 (K-101 VFD speed) · AO · DCS	Flow, multi-point averaging pitot in the duct (typical)	0.0–250.0	kNm³/h	176.8	L 150 · H 205	—	VFD: stops on trip	—	CC-PID-101
AI-101	CO2 in the flue gas to T-101 (wet basis); design 4.0, range 3–6	AIT-101 · AI · DCS	CO2 analyzer, NDIR, extractive (typical)	2.00–7.00	mol%	4.00	L 3.2 · H 5	—	—	—	CC-PID-101 (as AIT-101)
TI-101	Flue gas after the direct-contact cooler (battery limit)	TT-101 · AI · DCS	RTD Pt100 in a thermowell + transmitter	30.0–60.0	°C	40.0	—	—	—	—	CC-PID-101
AY-102	DCS calc: % of flooding in T-101 (GPDC, Fig 5-63)	— (DCS calculation)	DCS calculation (soft tag, no I/O)	0.0–120.0	%	62.1	H 80	—	—	—	CC-PID-101
PDI-102	T-101 pressure drop (packing, wash, distributors, demister)	PDT-102 · AI · DCS	Differential-pressure transmitter	0.00–15.00	kPa	4.00	H 6	—	—	—	CC-PID-101
TI-103	T-101 bed 1 top	TT-103 · AI · DCS	RTD Pt100 in a thermowell + transmitter	30.0–80.0	°C	52.9	—	—	—	—	CC-PID-101
TI-104	T-101 bed 1 middle	TT-104 · AI · DCS	RTD Pt100 in a thermowell + transmitter	30.0–80.0	°C	57.3	—	—	—	—	CC-PID-101
TI-105	T-101 bed 1 bottom	TT-105 · AI · DCS	RTD Pt100 in a thermowell + transmitter	30.0–80.0	°C	53.9	—	—	—	—	CC-PID-101
TI-106	T-101 bed 2 middle	TT-106 · AI · DCS	RTD Pt100 in a thermowell + transmitter	30.0–80.0	°C	48.5	—	—	—	—	CC-PID-101
TI-107	T-101 bed 2 bottom	TT-107 · AI · DCS	RTD Pt100 in a thermowell + transmitter	30.0–80.0	°C	43.6	—	—	—	—	CC-PID-101
LIC-109	T-101 sump level → FV-109 rich MEA (P-101, fail closed)	LT-109 · AI · DCS FV-109 · AO · DCS	Level, DP with remote seals (typical; foam-tolerant)	0.0–100.0	%	50.0	L 30 · H 70	—	FV-109 FC	—	CC-PID-101
TI-108	Rich solvent in the T-101 sump	TT-108 · AI · DCS	RTD Pt100 in a thermowell + transmitter	30.0–70.0	°C	43.6	—	—	—	—	CC-PID-101
LSHH-109	Independent sump high-high switch: 90 % trips I-1 (liquid into the gas inlet)	LT-109B (LS-109) · AI · SIS	Level, DP with remote seals (typical; foam-tolerant)	0–100	%	50	—	HH 90 (I-1)	—	SIL 1 (SIF-1)	CC-PID-101
LSLL-109	Sump low-low switch: 5 % stops P-101 (I-3)	LT-109B (LS-109) · AI · SIS	Level, DP with remote seals (typical; foam-tolerant)	0–100	%	50	—	LL 5 (I-3)	—	SIL 1 (SIF-5)	CC-PID-101
TIC-110	Treated gas out of the water wash → TV-110 cooling water to E-105 (fail open)	TT-110 · AI · DCS TV-110 · AO · DCS	RTD Pt100 in a thermowell + transmitter	20.0–80.0	°C	40.0	H 45	—	TV-110 FO	—	CC-PID-101
AI-111	CO2 in the treated gas to the stack	AIT-111 · AI · DCS	CO2 analyzer, NDIR, extractive (typical)	0.00–4.00	mol%	0.414	H 0.55	—	—	—	CC-PID-101 (as AIT-111)
AY-112	DCS calc: CO2 capture from AI-101 and AI-111 (inert basis); target ≥ 90 %	— (DCS calculation)	DCS calculation (soft tag, no I/O)	0.0–100.0	%	90.0	L 88	—	—	—	CC-PID-101
AI-113	Rich loading, mol CO2 / mol MEA (on-line analyzer)	AIT-113 · AI · DCS	CO2 loading analyzer, on-line (typical, TBC)	0.20–0.55	mol/mol	0.471	H 0.485	—	—	—	CC-PID-101 (as AIT-113)
FIC-201	Lean MEA to the absorber top (FV-201, fail open)	FT-201 · AI · DCS FV-201 · AO · DCS	Flow, magnetic flowmeter (typical)	0.0–500.0	t/h	269.7	L 220 · H 330	—	FV-201 FO	—	CC-PID-101
TIC-202	Lean MEA to T-101 after cooler E-104 → TV-202 cooling water (fail open)	TT-202 · AI · DCS TV-202 · AO · DCS	RTD Pt100 in a thermowell + transmitter	20.0–80.0	°C	40.0	H 45	—	TV-202 FO	—	CC-PID-101
AI-203	Lean loading, mol CO2 / mol MEA (on-line analyzer)	AIT-203 · AI · DCS	CO2 loading analyzer, on-line (typical, TBC)	0.10–0.45	mol/mol	0.246	H 0.28	—	—	—	CC-PID-101 (as AIT-203)
TI-204	Rich solvent to T-102 (E-101 cold outlet)	TT-204 · AI · DCS	RTD Pt100 in a thermowell + transmitter	40.0–130.0	°C	109.0	—	—	—	—	CC-PID-101

Tag	Service	Field device · I/O · system	Type (typical, TBC)	Range	Units	Normal	Alarm	Trip	Fail position	SIL	P&ID
TI-205	Lean solvent after E-101 (to cooler E-104)	TT-205 · AI · DCS	RTD Pt100 in a thermowell + transmitter	30.0–100.0	°C	51.6	—	—	—	—	CC-PID-101
FI-109	Rich solvent to the stripper	FT-109 · AI · DCS	Flow, magnetic flowmeter (typical)	0.0–500.0	t/h	282.2	—	—	—	—	CC-PID-101
TIC-206	Reboiler / T-102 sump temperature → FIC-207 steam set point (cascade)	TT-206 · AI · DCS	RTD Pt100 in a thermowell + transmitter	80.0–140.0	°C	120.0	L 117 · H 123	—	via FIC-207 (FV-207 FC)	—	CC-PID-101
TSHH-206	Reboiler high-high 130 °C: MEA thermal degradation, closes the steam (I-2)	TT-206B (TS-206) · AI · SIS	RTD Pt100 in a thermowell + transmitter	80–140	°C	120	—	HH 130 (I-2)	—	SIL 1 (SIF-3)	CC-PID-101
FIC-207	LP steam to reboiler E-102 (FV-207, fail closed)	FT-207 · AI · DCS FV-207 · AO · DCS	Flow, vortex on LP steam (typical)	0.00–40.00	t/h	22.94	—	—	FV-207 FC	—	CC-PID-101
AY-212	DCS calc: specific reboiler duty = steam × λ / CO2 product	— (DCS calculation)	DCS calculation (soft tag, no I/O)	2.00–6.00	GJ/t	3.94	H 4.3	—	—	—	CC-PID-101
LIC-208	T-102 sump level → FV-208 make-up water (fail closed)	LT-208 · AI · DCS FV-208 · AO · DCS	Level, DP with remote seals (typical; foam-tolerant)	0.0–100.0	%	50.0	L 30 · H 70	—	FV-208 FC	—	CC-PID-101
LSLL-208	T-102 sump low-low: 10 % closes the steam (I-2)	LT-208B (LS-208) · AI · SIS	Level, DP with remote seals (typical; foam-tolerant)	0–100	%	50	—	LL 10 (I-2 / I-3)	—	SIL 1 (SIF-4a); SIL 1 (SIF-4b)	CC-PID-101
PDI-209	T-102 packing pressure drop	PDT-209 · AI · DCS	Differential-pressure transmitter	0.00–8.00	kPa	1.49	—	—	—	—	CC-PID-101
AY-210	DCS calc: % of flooding in T-102 (GPDC, Fig 5-62)	— (DCS calculation)	DCS calculation (soft tag, no I/O)	0.0–120.0	%	50.6	H 80	—	—	—	CC-PID-101
TI-211	T-102 top temperature (stage 1)	TT-211 · AI · DCS	RTD Pt100 in a thermowell + transmitter	60.0–130.0	°C	103.1	—	—	—	—	CC-PID-101
PSHH-301	Stripper high-high pressure 250 kPa a closes the steam (I-2); PSV-301 at 350 kPa a	PT-301B (PS-301) · AI · SIS	Pressure transmitter	100–300	kPa a	175	—	HH 250 (I-2)	—	SIL 1 (SIF-2); SIL 1 (SIF-2R, proposed)	CC-PID-101
PIC-301	Reflux drum V-101 pressure → PV-301 CO2 product to compression (fail open)	PT-301 · AI · DCS PV-301 · AO · DCS	Pressure transmitter	100.0–300.0	kPa a	175.0	L 165 · H 190	—	PV-301 FO	—	CC-PID-101
TIC-302	Condenser E-103 outlet → TV-302 cooling water (fail open)	TT-302 · AI · DCS TV-302 · AO · DCS	RTD Pt100 in a thermowell + transmitter	20.0–100.0	°C	40.0	H 50	—	TV-302 FO	—	CC-PID-101
LIC-303	Reflux drum V-101 level → FV-303 reflux to T-102 (P-103, fail closed)	LT-303 · AI · DCS FV-303 · AO · DCS	Level, guided-wave radar (typical)	0.0–100.0	%	50.0	L 25 · H 75	—	FV-303 FC	—	CC-PID-101
FI-304	CO2 product to compression (dry basis)	FT-304 · AI · DCS	Flow, vortex on wet CO2, dry basis in the DCS (typical)	0.00–20.00	t/h	12.50	—	—	—	—	CC-PID-101
AY-214	DCS calc: capture from the CO2 product FI-304 / CO2 in the flue gas	— (DCS calculation)	DCS calculation (soft tag, no I/O)	0.0–120.0	%	90.0	—	—	—	—	CC-PID-101
AIT-213	MEA concentration (CO2-free basis, density/conductivity)	AIT-213 · AI · DCS	MEA concentration, density + conductivity (typical, TBC)	20.00–40.00	wt%	30.01	L 28 · H 32	—	—	—	CC-PID-101
JI-101	Blower K-101 shaft power (∝ flow × ΔP)	JT-101 · AI · DCS	Shaft power from the K-101 VFD (hard-wired AI, TBC)	0–800	kW	313	—	—	—	—	CC-PID-101
FI-208	Make-up water to T-102	FT-208 · AI · DCS	Flow, magnetic flowmeter (typical)	0.00–1.00	t/h	0.25	—	—	—	—	not on CC-PID-101
FI-303	Reflux water to T-102	FT-303 · AI · DCS	Flow, magnetic flowmeter (typical)	0.00–15.00	t/h	5.54	—	—	—	—	not on CC-PID-101
TI-401	Cooling-water supply	TT-401 · AI · DCS	RTD Pt100 in a thermowell + transmitter	20.0–45.0	°C	30.0	—	—	—	—	not on CC-PID-101

3 Final elements and motors

Tag	Service	Signals · I/O · system	Fail / trip action	SIF
XV-101	Flue-gas damper to T-101 (I-1)	XY-101 solenoid · DO · SIS ZSO-101 open · DI · SIS ZSC-101 closed · DI · SIS	FC (closes on air / power loss)	SIF-1
XV-207	LP steam shut-off to E-102 (I-1, I-2)	XY-207 solenoid · DO · SIS ZSO-207 open · DI · SIS ZSC-207 closed · DI · SIS	FC	SIF-2, SIF-3, SIF-4a
K-101	Flue gas blower motor (VFD)	HS-101 start/stop · DO · DCS run status · DI · DCS SI-101 speed feedback · AI · DCS I-1 trip to VFD · DO · SIS	stops	SIF-1
P-101	Rich amine pump motor	start/stop · DO · DCS run status · DI · DCS I-3 trip (LSLL-109) · DO · SIS	stops	SIF-5
P-102	Lean amine pump motor	start/stop · DO · DCS run status · DI · DCS I-3 trip (LSLL-208) · DO · SIS	stops	SIF-4b
P-103	Reflux pump motor	start/stop · DO · DCS run status · DI · DCS	stops	—
P-104	Wash water pump motor	start/stop · DO · DCS run status · DI · DCS	stops	—
HS-ESD	Emergency shutdown pushbutton (I-1)	HS-ESD · DI · SIS	—	manual

4 I/O count

System	AI	AO	DI	DO	Total	With 20 % spare
DCS	35	10	5	5	**55**	66
SIS	4	0	5	5	**14**	17
Total	39	10	10	10	**69**	

Counted from the field devices of part 2 and the signals of part 3. The shared SIS transmitter LT-109B is counted once. DCS calculations use no I/O. The SIS has its own I/O cards, separate from the DCS.

5 Control-valve check

The model gives each valve a capacity at 100 % open and a linear characteristic. With the pressure drop listed (assumed except PV-301), the capacity is a rated C_v. The design flow gives the C_v required, and their ratio is the opening at design, which must equal the opening the model holds there. Typical practice puts a control valve between 20 and 80 % open at normal flow.

Valve	Loop	Fluid	Design t/h	Capacity 100 % t/h	ρ kg/m³	ΔP kPa	C_v design	C_v rated	Opening linear / model	Eq. % (R 50)	20–80 %	Note
FV-109	LIC-109	Rich MEA (P-101 discharge)	282.19	545.0	1,112	100 (assumed)	309.5	597.7	51.8 / 51.8 %	83.2 %	✓	
FV-201	FIC-201	Lean MEA to T-101	269.70	520.0	1,064	100 (assumed)	302.4	583.0	51.9 / 51.9 %	83.2 %	✓	
TV-202	TIC-202	Cooling water to E-104	1,830.77	3,600.0	995	70 (assumed)	2,537.2	4,989.0	50.9 / 50.9 %	82.7 %	✓	CW rise 1.5 K for 3.13 MW
TV-110	TIC-110	Cooling water to E-105	896.45	1,800.0	995	70 (assumed)	1,242.3	2,494.5	49.8 / 49.8 %	82.2 %	✓	CW rise 5.5 K for 5.77 MW
FV-207	FIC-207	LP steam to E-102	22.94	47.0	1.84	50 (assumed)	94.4	193.3	48.8 / 48.8 %	81.7 %	✓	
FV-208	LIC-208	Demin make-up water	0.25	5.0	994	150 (assumed)	0.2	4.7	4.9 / 4.9 %	23.1 %	✗	
PV-301	PIC-301	Wet CO2 product	12.73	26.7	2.88	25 (model)	59.3	124.5	47.6 / 47.6 %	81.0 %	✓	P1 175 → P2 150 kPa a (model K.Pdown)
TV-302	TIC-302	Cooling water to E-103	795.28	1,440.0	995	70 (assumed)	1,102.1	1,995.6	55.2 / 55.2 %	84.8 %	✓	CW rise 4.4 K for 4.08 MW
FV-303	LIC-303	Reflux water (P-103)	5.54	10.8	992	100 (assumed)	6.4	12.5	51.3 / 51.3 %	82.9 %	✓	

5.1 Worked calculation · FV-201 (liquid, IEC 60534-2-1)

Q = 269.7 t/h / 1,064.1 kg/m³ = 253.5 m³/h; G = 1.065; ΔP = 100 kPa = 1.00 bar (assumed)

$C_v = 1.156\ Q\ \sqrt{(G/\Delta P)} = 1.156 \times 253.5 \times \sqrt{(1.065 / 1.00)}$ = **302.4**; rated (model capacity 520 t/h) C_v = **583.0**

Opening at design = 302.4 / 583.0 = **51.9 %** (model 51.9 %); equal-percentage trim: $1 + \ln(C_v/C_{v,rated})/\ln 50$ = 83.2 %

6 Alarm and trip hierarchy

Trip instrument	SIS transmitter	Trip	Control loop	Loop alarm on the same side	Range (SIS / loop)	Interlock	SIL
LSHH-109	LT-109B (LS-109)	HH 90 (I-1)	LIC-109	H 70 %	0–100 / 0.0–100.0	I-1	SIL 1 (SIF-1)
LSLL-109	LT-109B (LS-109)	LL 5 (I-3)	LIC-109	L 30 %	0–100 / 0.0–100.0	I-3	SIL 1 (SIF-5)
TSHH-206	TT-206B (TS-206)	HH 130 (I-2)	TIC-206	H 123 °C	80–140 / 80.0–140.0	I-2	SIL 1 (SIF-3)
LSLL-208	LT-208B (LS-208)	LL 10 (I-2 / I-3)	LIC-208	L 30 %	0–100 / 0.0–100.0	I-2 / I-3	SIL 1 (SIF-4a); SIL 1 (SIF-4b)
PSHH-301	PT-301B (PS-301)	HH 250 (I-2)	PIC-301	H 190 kPa a	100–300 / 100.0–300.0	I-2	SIL 1 (SIF-2); SIL 1 (SIF-2R, proposed)

Each trip lies beyond the alarm on the same side of its control loop, so the operator is warned before the SIS acts. Each trip setting lies inside the range of its own SIS transmitter. The pressure transmitters PT-301 and PT-301B end at 300 kPa a, below the PSV-301 set of 350 kPa a (MEA-PE-RLB-001), see EQ-I06.

7 Where the trips and SIFs are specified

- What each trip and alarm does: cause-and-effect matrix MEA-IN-CE-001 (not repeated here).
- SIL, voting, response time, proof-test interval and process safety time of each SIF: MEA-IN-SRS-001 (the SIL column of part 2 is copied from it).
- PSV set and relief cases: relief load basis MEA-PE-RLB-001.

8 Engineering queries

EQ	Subject	Documents that disagree	What	Suggested resolution
EQ-I01	FIC-201 span	makeCtrls() FIC-201 0–500 t/h; DCS trend range CRANGE 0–400 t/h	The controller works on 0–500 t/h but the trend and faceplate scale is 0–400 t/h. A transmitter calibrated to one and a display scaled to the other read the same mA as two different flows.	Calibrate FT-201 0–500 t/h (it must cover the FV-201 capacity) and set the DCS range to the same.
EQ-I02	Analyzer tag letters	Screen bubbles AI-101/111/113/203; CC-PID-101 AIT-101/111/113/203; AIT-213 on both	One instrument carries two tags in two documents.	Use AIT-xxx for the transmitter and AI-xxx (or AIC) for the DCS indication; update the screen or the P&ID.
EQ-I03	Tags not on the P&ID	Screen / IND: FI-208, FI-303, TI-401; AY-212, AY-214 only in note 8	Instruments that the DCS shows have no bubble on CC-PID-101.	Add FT-208, FT-303 and TT-401 to CC-PID-101 at the next revision.
EQ-I04	FV-208 oversized; FI-208 range	Model K.FmaxMU 5 t/h, FV-208 at about 5 % open at design; IND FI-208 0–1 t/h	The make-up valve works at the bottom of its stroke, where control is poor, and the flowmeter cannot read the flow when the valve is wide open.	Size FV-208 for about 1 t/h (design opening 50–70 %) or add a small trim; re-range FI-208 to cover the valve capacity.
EQ-I05	Cooling-water flows in the model	K.Wcwl / Wcww / Wcwc (CW valve capacities) vs exchanger duties	At design the CW temperature rise is only 1.5–5.5 K (E-104 1.5 K), so the CW valve Cv in this list is much larger than a real CW balance (8–10 K rise, typical) would give.	Size TV-202 / TV-110 / TV-302 from the CW balance of the utility study; keep the model values for training only.
EQ-I06	Pressure ranges vs the PSV set	IND PSHH-301 and PIC-301 0/100–300 kPa a; MEA-PE-RLB-001 PSV-301 set 350, relieving 375 kPa a	Neither PT-301 nor PT-301B can show the approach to the PSV set or the relieving pressure; they peg at 300 kPa a.	Range PT-301 and PT-301B 0–400 kPa a.
EQ-I07	Steam valve ΔP	Model (reboiler condenses at the header pressure); MEA-PE-RLB-001 EQ-R08	FV-207 is sized here with an assumed 50 kPa drop, which the model does not have.	Confirm the LP header pressure at E-102 and recompute FV-207.
EQ-I08	Flow ranges below valve capacity	IND FI-109 0–500 vs FV-109 545 t/h; FIC-207 0–40 vs FV-207 47 t/h	A wide-open valve drives the transmitter over range, so the operator loses the reading in the upset where it matters.	Range each flow transmitter to at least the valve capacity, or record the reason (e.g. pump curve limit).